

SSTO

SSTO $\Rightarrow$ Single Stage To Orbit

$\rightarrow$ no hardware dropped during flight (Hayman)

Difficulty: - almost complete mass is to be taken to the orbit and so room for payload is very less.

Tsiolkovsky Rocket Equation.

$$\Delta v = I_{sp} \times g_0 \times \ln \left(\frac{m_0}{m_f} \right)$$

where, Δv = velocity from given amount of propellant

I_{sp} = specific impulse (Hayman: engine efficiency in sec)

g_0 = standard gravity (9.80665 ms^{-2} for earth)
 $\hookrightarrow$ field strength

m_0 = initial mass (payload + hardware/structure) + propellant)

$$= m_p + m_s + m_{pr} \quad \dots \textcircled{a}$$

m_f = final mass (payload + structure)

$$= m_p + m_s \quad \dots \textcircled{b}$$

* Facts: to reach LEO, $\Delta v \approx 9.4 \text{ km/s}$ (typical value keeping in mind gravity ~~and~~, drag losses and orbital velocity ($\sim 7.8 \text{ km/s}$))

LEO $\sim 160 \text{ km}$ to 2000 km

Now, Structural Coefficient (ϵ) is the ratio of structural mass to the dry mass (mass without that payload or $m_s + m_{pr}$)

$$\text{so, } \epsilon = \frac{m_s}{m_s + m_{pr}} \quad \dots \textcircled{1}$$

more ϵ , heavier tanks. (proportional to the amount of propellant held)

$$\text{so, } m_s = \frac{\epsilon}{1-\epsilon} m_{pr} \dots (2)$$

substituting (2) in (a) and (b),

$$m_o = m_p + \frac{\epsilon}{1-\epsilon} m_{pr} + m_{pr} = m_p + \frac{1}{1-\epsilon} m_{pr} \dots (3)$$

$$m_f = m_p + \frac{\epsilon}{1-\epsilon} m_{pr} \dots (4)$$

New payload fraction (λ) = m_p / m_o

$$m_o = m_p + \frac{1}{1-\epsilon} m_{pr}$$

$$\text{let } m_{pr} / m_o = \mu$$

$$\text{so, } 1 = \lambda + \left(\frac{1}{1-\epsilon} \right) \mu$$

$$\Rightarrow \mu = (1-\lambda)(1-\epsilon)$$

$$\text{Now, mass fraction, } \frac{m_f}{m_o} = \lambda + \frac{\epsilon}{1-\epsilon} \mu$$

$$= \lambda + \epsilon(1-\lambda)$$

$$m_f / m_o = \epsilon + \lambda(1-\epsilon)$$

$$\text{But, By rocket equation, } \frac{m_o}{m_f} = e^{4v/I_{sp} g_0}$$

$$\text{let } e^{4v/I_{sp} g_0} = R$$

$$\text{so, } \frac{m_o}{m_f} = \frac{1}{R}$$

$$\text{so, } \epsilon + \lambda(1 - \epsilon) = e^{-\Delta v / I_{sp} g_0} g_0$$

$$\lambda = \frac{e^{-\Delta v / I_{sp} g_0} - \epsilon}{1 - \epsilon}$$

Higher λ , higher payload capacity.

general cases:-

① for chemical SSTO,

$$\Delta v = 9400 \text{ m s}^{-1}$$

$$I_{sp} = 450 \text{ s}$$

$$\epsilon = 0.10$$

$$\text{so, } \lambda = \frac{e^{-(9400 / (450 \times 9.80665))} - 0.10}{1 - 0.10} = 0.0209$$

$$\lambda \approx 2.1\% \text{ (barely possible)}$$

② for nuclear SSTO (without taking into account mass of reactor)

$$\Delta v = 9400 \text{ m s}^{-1}; I_{sp} = 800 \text{ s}; \epsilon = 0.10$$

$$\lambda = \frac{e^{-(9400 / (800 \times 9.80665))} - 0.1}{1 - 0.1} = 0.22416$$

$$\lambda \approx 22.4\% \text{ (very high)}$$

③ for nuclear SSTO (with mass of reactor in account)

$$\Delta v = 9400 \text{ m s}^{-1}, I_{sp} = 800 \text{ s}; \epsilon = 0.2$$

$$\lambda = \frac{e^{-(9400 / (800 \times 9.80665))} - 0.2}{1 - 0.2} = 0.127183$$

$$\lambda \approx 12.7\% \text{ (lower than the case above)}$$

— x —
Fin.